

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. ADT fields = non-recursive fields annotated with @BENCH adt_fields; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup >1× means SoA is faster; **bold** marks >1.1×.

Program	ADT	SoA	Fold passes			Map passes		
	fields	bufs	AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
Compiler	9	8	0.1235	0.0389	3.18 ×	0.1584	0.1772	0.89×
DBQuery	15	13	0.1185	0.0881	1.35 ×	0.1365	0.1655	0.83×
DecisionTree	7	6	3.637	2.934	1.24 ×	—	—	—
DomTree	15	14	0.1362	0.0606	2.25 ×	0.0583	0.0786	0.74×
KDTree	17	16	0.1746	0.1402	1.25 ×	—	—	—
LinearListReduction	11	11	0.0292	0.0050	5.83 ×	—	—	—
List	2	2	0.0882	0.0895	0.99×	0.2971	0.3619	0.82×
MonoTree	3	2	0.0171	0.0224	0.76×	0.0455	0.0428	1.06×
ObjectGraph	5	4	0.0893	0.0798	1.12 ×	0.1437	0.1612	0.89×
OctTree	16	9	0.0345	0.0310	1.12 ×	0.0387	0.0439	0.88×
PiecewiseFunctions	8	7	0.1350	0.0913	1.48 ×	0.2376	0.2586	0.92×
TernaryTree	5	3	0.0275	0.0258	1.07×	0.0832	0.0777	1.07×
Trie	10	9	0.0621	0.0314	1.98 ×	0.1492	0.1652	0.90×
ColorOctree	25	18	0.0874	0.0768	1.14 ×	—	—	—

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. >1× means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
Compiler	1.59 ×	0.52×	4.14 ×	0.80×	14.17 ×	0.37×	6.72 ×
DBQuery	1.05×	0.47×	1.31 ×	0.46×	8.32 ×	0.27×	6.24 ×
DecisionTree	1.24 ×	0.59×	17.81 ×	2.36 ×	0.97×	2.52 ×	36.81 ×
DomTree	1.53 ×	0.57×	2.80 ×	1.07×	21.53 ×	0.16×	10.54 ×
KDTree	1.24 ×	0.77×	1.51 ×	0.78×	0.19×	0.79×	4.63 ×
LinearListReduction	5.88 ×	0.90×	32.36 ×	7.60 ×	41.32 ×	7.57 ×	11.63 ×
List	0.81×	0.80×	3.30 ×	0.76×	0.11×	0.67×	1.45 ×
MonoTree	0.96×	0.85×	0.55×	0.85×	0.24×	0.71×	1.22 ×
ObjectGraph	0.96×	0.75×	6.54 ×	0.77×	0.54×	0.71×	2.88 ×
OctTree	0.99×	0.73×	0.77×	0.56×	0.21×	0.53×	4.09 ×
PiecewiseFunctions	1.11 ×	0.60×	2.78 ×	0.67×	2.17 ×	0.41×	4.46 ×
TernaryTree	1.14 ×	0.79×	0.39×	0.45×	0.22×	0.79×	1.38 ×
Trie	1.13 ×	0.56×	3.66 ×	0.88×	3.34 ×	0.18×	6.43 ×
ColorOctree	1.14 ×	0.84×	4.58 ×	1.14 ×	0.36×	1.16 ×	4.46 ×
Geomean	1.27 ×	0.68×	2.84 ×	0.95×	1.29 ×	0.66×	4.78 ×

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Compiler	9.3e+08/ 5.8e+08	9.9e+08 /1.9e+09	1.3e+07/ 3.1e+06	2.2e+05 /2.7e+05	1.2e+06/ 8.7e+04	1.4e+04 /3.7e+04	5.1e+07/ 7.6e+06
DBQuery	9.3e+08/ 8.8e+08	1.1e+09 /2.2e+09	3.9e+06/ 3.0e+06	1.6e+05 /3.6e+05	8.1e+05/ 9.7e+04	9.8e+03 /3.7e+04	2.1e+07/ 3.3e+06
DecisionTree	1.8e+10/ 1.5e+10	4.0e+10 /6.8e+10	5.4e+08/ 3.0e+07	1.1e+05/ 4.7e+04	2.0e+06 /2.1e+06	1.1e+05/ 4.4e+04	1.1e+09/ 2.9e+07
DomTree	8.1e+08/ 5.3e+08	1.2e+09 /2.2e+09	1.2e+07/ 4.2e+06	1.1e+05/ 1.1e+05	1.1e+06/ 5.0e+04	4.6e+03 /3.0e+04	5.7e+07/ 5.4e+06
KDTree	8.8e+08/ 7.1e+08	3.1e+09 /4.0e+09	1.5e+07/ 9.6e+06	8.2e+02 /1.0e+03	2.3e+04 /1.2e+05	8.2e+02 /1.0e+03	4.4e+07/ 9.6e+06
LinearListReduction	1.5e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.5e+06/ 4.7e+04	3.6e+02/ 4.7e+01	2.8e+05/ 6.9e+03	3.6e+02/ 4.7e+01	1.2e+07/ 1.1e+06
List	1.2e+09 /1.5e+09	4.5e+09 /5.6e+09	9.4e+06/ 2.8e+06	4.1e+05 /5.4e+05	2.3e+04 /2.0e+05	6.7e+03 /1.0e+04	4.8e+07/ 3.3e+07
MonoTree	2.5e+08 /2.6e+08	9.1e+08 /1.1e+09	2.6e+05 /4.7e+05	4.0e+04 /4.6e+04	3.2e+03 /1.3e+04	1.5e+03 /2.2e+03	2.0e+06/ 1.7e+06
ObjectGraph	8.4e+08 /8.7e+08	3.0e+09 /4.0e+09	9.5e+06/ 1.5e+06	1.9e+05 /2.4e+05	2.9e+04 /5.3e+04	1.1e+04 /1.6e+04	2.4e+07/ 8.2e+06
OctTree	2.6e+08 /2.7e+08	8.2e+08 /1.1e+09	6.6e+05 /8.6e+05	6.1e+04 /1.1e+05	4.7e+03 /2.2e+04	1.4e+03 /2.6e+03	4.5e+06/ 1.1e+06
PiecewiseFunctions	1.2e+09/ 1.1e+09	2.8e+09 /4.7e+09	1.4e+07/ 5.1e+06	3.2e+05 /4.8e+05	2.0e+05/ 9.0e+04	2.2e+04 /5.3e+04	4.5e+07/ 1.0e+07
TernaryTree	4.1e+08/ 3.6e+08	1.2e+09 /1.5e+09	5.6e+05 /1.4e+06	6.5e+04 /1.5e+05	4.4e+03 /2.0e+04	2.0e+03 /2.6e+03	3.3e+06/ 2.4e+06
Trie	6.4e+08/ 5.7e+08	1.2e+09 /2.2e+09	9.0e+06/ 2.5e+06	2.0e+05 /2.3e+05	1.5e+05/ 4.6e+04	7.2e+03 /4.0e+04	2.6e+07/ 4.0e+06
ColorOctree	4.4e+08/ 3.9e+08	1.6e+09 /2.0e+09	9.5e+06/ 2.1e+06	4.6e+02/ 4.0e+02	2.7e+04 /7.3e+04	4.7e+02/ 4.0e+02	1.9e+07/ 4.2e+06

Table 4: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
Compiler	0.2819	1.107	0.2161	<i>OOM</i>
DBQuery	0.2550	0.3259	0.2535	1.122
DecisionTree	3.637	5.618	2.934	24.84
DomTree	0.1945	0.3337	0.1392	0.5255
KDTree	0.1746	0.2428	0.1402	0.7257
LinearListReduction	0.0292	0.1382	0.0050	<i>OOM</i>
List	0.3853	<i>OOM</i>	0.4514	<i>OOM</i>
MonoTree	0.0626	0.1412	0.0652	0.2738
ObjectGraph	0.2330	0.4790	0.2410	0.7039
OctTree	0.0733	0.0804	0.0749	<i>OOM</i>
PiecewiseFunctions	0.3726	0.5529	0.3499	2.136
TernaryTree	0.1108	0.1438	0.1035	0.6103
Trie	0.2113	0.2925	0.1966	1.005
ColorOctree	0.0874	0.1178	0.0768	<i>OOM</i>

Table 5: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. >1× means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Compiler	1.30 ×	3.92 ×	5.12 ×	—
DBQuery	1.01×	1.28 ×	1.29 ×	0.29×
DecisionTree	1.24 ×	1.54 ×	1.91 ×	0.23×
DomTree	1.40 ×	1.72 ×	2.40 ×	0.63×
KDTree	1.25 ×	1.39 ×	1.73 ×	0.33×
LinearListReduction	5.83 ×	4.74 ×	27.62 ×	—
List	0.85×	—	—	—
MonoTree	0.96×	2.26 ×	2.17 ×	0.52×
ObjectGraph	0.97×	2.06 ×	1.99 ×	0.68×
OctTree	0.98×	1.10×	1.07×	—
PiecewiseFunctions	1.06×	1.48 ×	1.58 ×	0.26×
TernaryTree	1.07×	1.30 ×	1.39 ×	0.24×
Trie	1.07×	1.38 ×	1.49 ×	0.29×
ColorOctree	1.14 ×	1.35 ×	1.53 ×	—

Table 6: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
blockCountPass	F	2/9	78%	0.0117 $\pm$ 0.00001	0.0809 $\pm$ 0.00120	0.0019$\pm$0.00000	<i>OOM</i>	6.11$\times$	6.90$\times$	42.17$\times$	–
branchStatsPass	F	2/9	78%	0.0114 $\pm$ 0.00001	0.0814 $\pm$ 0.00002	0.0035$\pm$0.00000	<i>OOM</i>	3.21$\times$	7.14$\times$	22.93$\times$	–
castInstCountPass	F	3/9	67%	0.0177 $\pm$ 0.00000	0.0814 $\pm$ 0.00004	0.0013$\pm$0.00000	<i>OOM</i>	13.67$\times$	4.61$\times$	63.09$\times$	–
has cycle	F	4/9	56%	0.0240 $\pm$ 0.00011	0.0929 $\pm$ 0.00117	0.0171$\pm$0.00015	<i>OOM</i>	1.41$\times$	3.87$\times$	5.45$\times$	–
instCountPass	F	2/9	78%	0.0117 $\pm$ 0.00000	0.0815 $\pm$ 0.00007	0.0019$\pm$0.00000	<i>OOM</i>	6.02$\times$	6.95$\times$	41.85$\times$	–
latencyModelPass	F	3/9	67%	0.0117 $\pm$ 0.00001	0.0815 $\pm$ 0.00084	0.0029$\pm$0.00000	<i>OOM</i>	3.97$\times$	6.97$\times$	27.68$\times$	–
memoryOpStatsPass	F	3/9	67%	0.0115 $\pm$ 0.00001	0.0814 $\pm$ 0.00082	0.0038$\pm$0.00001	<i>OOM</i>	3.03$\times$	7.07$\times$	21.41$\times$	–
stripSideEffectsPass	M	7/9	22%	0.0793$\pm$0.00119	0.1877 $\pm$ 0.00115	0.0897 $\pm$ 0.00112	<i>OOM</i>	0.88 $\times$	2.37$\times$	2.09$\times$	–
targetReturnPass	M	9/9	0%	0.0791$\pm$0.00146	0.1859 $\pm$ 0.00191	0.0874 $\pm$ 0.00010	<i>OOM</i>	0.90 $\times$	2.35$\times$	2.13$\times$	–
throughputModelPass	F	3/9	67%	0.0117 $\pm$ 0.00001	0.0815 $\pm$ 0.00131	0.0030$\pm$0.00000	<i>OOM</i>	3.96$\times$	6.97$\times$	27.57$\times$	–
verifyIR	F	9/9	0%	0.0121 $\pm$ 0.00006	0.0703 $\pm$ 0.00079	0.0034$\pm$0.00038	<i>OOM</i>	3.55$\times$	5.79$\times$	20.59$\times$	–
Total				0.2819	1.107	0.2161	–	1.30$\times$	3.92$\times$	5.12$\times$	–
Geomean								3.15$\times$	5.15$\times$	16.25$\times$	–

Table 7: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	5.9e+07/ 9.7e+06	5.1e+07/ 4.5e+07	1.2e+06/ 6.2e+03	3.2e+02/ 2.3e+01	2.2e+05/ 1.1e+03	3.2e+02/ 2.3e+01	4.7e+06/ 2.2e+03
branchStatsPass	F	5.8e+07/ 1.8e+07	7.9e+07 /8.6e+07	8.2e+05/ 2.1e+04	2.4e+02/ 4.5e+01	4.9e+04/ 3.2e+03	2.4e+02/ 4.6e+01	4.6e+06/ 4.3e+05
castInstCountPass	F	8.9e+07/ 6.6e+06	5.1e+07/ 3.9e+07	1.3e+04 /2.2e+04	8.2e+01/ 1.6e+01	8.2e+02 /1.1e+03	8.5e+01/ 1.6e+01	4.6e+06/ 3.2e+03
has cycle	F	1.2e+08/ 8.6e+07	1.1e+08 /1.5e+08	2.3e+06/ 2.7e+05	4.0e+02/ 1.0e+02	1.5e+05/ 2.2e+04	4.0e+02/ 1.0e+02	5.1e+06/ 2.3e+06
instCountPass	F	6.0e+07/ 9.9e+06	5.6e+07 /5.7e+07	1.2e+06/ 1.5e+04	2.7e+02/ 3.4e+01	2.3e+05/ 1.1e+03	2.7e+02/ 3.5e+01	4.7e+06/ 3.0e+03
latencyModelPass	F	5.9e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.4e+04	3.0e+02/ 3.6e+01	2.0e+05/ 3.9e+03	3.0e+02/ 3.6e+01	4.7e+06/ 5.2e+05
memoryOpStatsPass	F	5.8e+07/ 1.9e+07	8.5e+07 /9.1e+07	6.7e+05/ 2.8e+04	2.8e+02/ 4.3e+01	3.0e+04/ 3.2e+03	2.8e+02/ 4.6e+01	4.6e+06/ 4.4e+05
stripSideEffectsPass	M	1.5e+08 /2.0e+08	1.8e+08 /6.0e+08	1.8e+06/ 1.2e+06	1.1e+05 /1.4e+05	7.0e+03 /2.0e+04	4.8e+03 /1.6e+04	4.1e+06/ 1.2e+06
targetReturnPass	M	1.5e+08 /1.9e+08	1.7e+08 /6.3e+08	1.4e+06/ 1.4e+06	1.1e+05 /1.3e+05	6.3e+03 /2.4e+04	6.4e+03 /2.0e+04	4.2e+06/ 1.6e+06
throughputModelPass	F	5.9e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.1e+04	3.4e+02/ 4.2e+01	2.0e+05/ 3.5e+03	3.4e+02/ 4.3e+01	4.7e+06/ 5.4e+05
verifyIR	F	6.2e+07/ 1.7e+07	7.4e+07 /8.6e+07	8.9e+05/ 1.8e+04	3.6e+02/ 1.2e+02	1.4e+05/ 4.0e+03	3.6e+02/ 1.2e+02	4.6e+06/ 4.5e+05
Total		9.3e+08/5.8e+08	9.9e+08/1.9e+09	1.3e+07/3.1e+06	2.2e+05/2.7e+05	1.2e+06/8.7e+04	1.4e+04/3.7e+04	5.1e+07/7.6e+06

Table 8: Per-pass PAPI counters for **Compiler** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
blockCountPass	F	4.1e+08/–	1.6e+08/–	2.9e+06/–	5.9e+02/–	1.7e+05/–	5.8e+02/–	1.4e+07/–
branchStatsPass	F	4.1e+08/–	1.8e+08/–	2.7e+06/–	7.3e+02/–	1.3e+05/–	7.3e+02/–	1.4e+07/–
castInstCountPass	F	4.1e+08/–	1.8e+08/–	2.7e+06/–	6.0e+02/–	1.3e+05/–	6.1e+02/–	1.4e+07/–
has cycle	F	4.7e+08/–	2.1e+08/–	2.5e+06/–	2.9e+02/–	7.0e+04/–	2.9e+02/–	1.7e+07/–
instCountPass	F	4.1e+08/–	1.6e+08/–	2.9e+06/–	4.6e+02/–	1.8e+05/–	4.6e+02/–	1.4e+07/–
latencyModelPass	F	4.1e+08/–	1.7e+08/–	2.6e+06/–	7.3e+02/–	1.3e+05/–	7.4e+02/–	1.4e+07/–
memoryOpStatsPass	F	4.1e+08/–	1.9e+08/–	2.6e+06/–	5.9e+02/–	1.2e+05/–	5.9e+02/–	1.4e+07/–
stripSideEffectsPass	M	6.9e+08/–	3.3e+08/–	1.8e+06/–	1.1e+05/–	1.7e+04/–	3.1e+04/–	2.3e+07/–
targetReturnPass	M	6.8e+08/–	3.2e+08/–	1.9e+06/–	1.0e+05/–	1.8e+04/–	3.6e+04/–	2.3e+07/–
throughputModelPass	F	4.1e+08/–	1.7e+08/–	2.7e+06/–	5.1e+02/–	1.3e+05/–	5.1e+02/–	1.4e+07/–
verifyIR	F	3.6e+08/–	1.2e+08/–	2.5e+06/–	4.5e+02/–	2.0e+05/–	4.6e+02/–	1.1e+07/–
Total		5.1e+09/–	2.2e+09/–	2.8e+07/–	2.2e+05/–	1.3e+06/–	7.2e+04/–	1.7e+08/–

Table 9: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearQueryFlags	M	14/15	7%	0.0685$\pm$0.00114	0.0755 $\pm$ 0.00008	0.0817 $\pm$ 0.00006	0.2996 $\pm$ 0.00131	0.84 $\times$	1.10$\times$	0.92 $\times$	0.25 $\times$
countJoins	F	3/15	80%	0.0185 $\pm$ 0.00001	0.0283 $\pm$ 0.00001	0.0112$\pm$0.00001	0.0854 $\pm$ 0.00156	1.65$\times$	1.53$\times$	2.52$\times$	0.33 $\times$
filterSelectivitySkew	F	5/15	67%	0.0185 $\pm$ 0.00001	0.0281 $\pm$ 0.00115	0.0122$\pm$0.00072	0.0869 $\pm$ 0.00155	1.52$\times$	1.51$\times$	2.30$\times$	0.32 $\times$
hashJoinPressure	F	5/15	67%	0.0189 $\pm$ 0.00083	0.0322 $\pm$ 0.00002	0.0125$\pm$0.00002	0.0864 $\pm$ 0.00002	1.51$\times$	1.71$\times$	2.58$\times$	0.37 $\times$
scaleCosts	M	15/15	0%	0.0680$\pm$0.00011	0.0756 $\pm$ 0.00139	0.0838 $\pm$ 0.00006	0.2983 $\pm$ 0.00129	0.81 $\times$	1.11$\times$	0.90 $\times$	0.25 $\times$
sumCost	F	6/15	60%	0.0185 $\pm$ 0.00009	0.0280 $\pm$ 0.00013	0.0149$\pm$0.00007	0.0886 $\pm$ 0.00028	1.24$\times$	1.52$\times$	1.89$\times$	0.32 $\times$
sumMemory	F	6/15	60%	0.0183 $\pm$ 0.00140	0.0283 $\pm$ 0.00001	0.0152$\pm$0.00002	0.0887 $\pm$ 0.00004	1.20$\times$	1.55$\times$	1.86$\times$	0.32 $\times$
sumRows	F	6/15	60%	0.0258 $\pm$ 0.00003	0.0297 $\pm$ 0.00002	0.0221$\pm$0.00001	0.0877 $\pm$ 0.00002	1.17$\times$	1.15$\times$	1.35$\times$	0.34 $\times$
Total				0.2550	0.3259	0.2535	1.122	1.01 $\times$	1.28$\times$	1.29$\times$	0.29 $\times$
Geomean								1.21$\times$	1.38$\times$	1.67$\times$	0.31 $\times$

Table 10: Per-pass PAPI counters for **DBQuery** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.7e+08 /2.1e+08	1.9e+08 /7.2e+08	2.9e+05 /1.2e+06	9.6e+04 /1.7e+05	3.7e+04/ 2.3e+04	4.4e+03 /1.5e+04	2.4e+06/ 9.3e+05
countJoins	F	9.4e+07/ 5.7e+07	7.4e+07/ 7.1e+07	6.1e+05/ 2.6e+04	4.4e+02/ 9.7e+01	1.4e+05/ 2.1e+03	4.4e+02/ 6.3e+01	2.6e+06/ 2.1e+03
filterSelectivitySkew	F	9.4e+07/ 6.2e+07	1.0e+08 /1.1e+08	5.2e+05/ 7.1e+04	2.2e+02/ 8.6e+01	1.2e+05/ 6.2e+03	2.2e+02/ 9.2e+01	2.6e+06/ 2.2e+05
hashJoinPressure	F	9.6e+07/ 6.3e+07	8.4e+07 /9.9e+07	7.3e+05/ 9.8e+04	3.5e+02/ 9.8e+01	1.9e+05/ 7.5e+03	3.5e+02/ 9.8e+01	2.7e+06/ 1.2e+05
scaleCosts	M	1.6e+08 /2.2e+08	2.2e+08 /7.5e+08	2.7e+05 /1.3e+06	6.6e+04 /1.8e+05	3.2e+04/ 2.8e+04	3.8e+03 /2.1e+04	2.4e+06/ 1.1e+06
sumCost	F	9.3e+07/ 7.5e+07	9.0e+07 /1.2e+08	5.5e+05/ 8.0e+04	1.0e+02/ 6.3e+01	1.2e+05/ 1.1e+04	1.1e+02/ 6.8e+01	2.7e+06/ 3.3e+05
sumMemory	F	9.3e+07/ 7.7e+07	9.0e+07 /1.2e+08	6.8e+05/ 8.1e+04	3.4e+02/ 5.8e+01	1.6e+05/ 1.1e+04	3.4e+02/ 6.0e+01	2.7e+06/ 3.2e+05
sumRows	F	1.3e+08/ 1.1e+08	2.1e+08 /2.3e+08	2.6e+05/ 1.0e+05	1.3e+02/ 1.1e+02	2.0e+04/ 7.8e+03	1.3e+02/ 1.1e+02	2.4e+06/ 2.9e+05
Total		9.3e+08/8.8e+08	1.1e+09/2.2e+09	3.9e+06/3.0e+06	1.6e+05/3.6e+05	8.1e+05/9.7e+04	9.8e+03/3.7e+04	2.1e+07/3.3e+06

Table 11: Per-pass PAPI counters for DBQuery (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearQueryFlags	M	2.0e+08 /1.3e+09	3.2e+08 /2.4e+09	6.1e+05 /2.0e+07	9.9e+04 /7.3e+05	1.9e+04 /4.8e+04	3.1e+03 /4.0e+04	2.3e+06/ 1.0e+06
countJoins	F	1.4e+08 /4.3e+08	1.6e+08 /7.7e+08	1.9e+06/ 3.1e+05	2.6e+02 /7.4e+02	1.1e+05/ 1.3e+04	2.6e+02/ 1.2e+02	2.3e+06/ 2.4e+04
filterSelectivitySkew	F	1.4e+08 /4.4e+08	2.0e+08 /7.9e+08	1.5e+06/ 8.3e+05	1.8e+02 /9.6e+02	6.3e+04/ 1.7e+04	1.8e+02 /3.3e+02	2.2e+06/ 1.8e+05
hashJoinPressure	F	1.6e+08 /4.4e+08	1.8e+08 /7.7e+08	1.7e+06/ 6.4e+05	1.5e+02 /8.4e+02	1.0e+05/ 1.6e+04	1.5e+02 /2.0e+02	2.2e+06/ 9.7e+04
scaleCosts	M	2.0e+08 /1.3e+09	3.5e+08 /2.4e+09	5.4e+05 /2.3e+07	7.2e+04 /3.5e+05	1.6e+04 /5.7e+04	3.6e+03 /5.0e+04	2.3e+06/ 1.2e+06
sumCost	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	1.7e+06/ 1.2e+06	1.3e+02 /8.6e+02	7.3e+04/ 2.2e+04	1.3e+02 /5.4e+02	2.3e+06/ 3.1e+05
sumMemory	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	1.5e+06/ 1.1e+06	2.6e+02 /8.8e+02	6.8e+04/ 2.1e+04	2.6e+02 /4.7e+02	2.2e+06/ 3.0e+05
sumRows	F	1.5e+08 /4.4e+08	2.3e+08 /8.3e+08	1.6e+06/ 1.2e+06	1.9e+02 /8.2e+02	5.6e+04/ 2.2e+04	1.9e+02 /4.8e+02	2.2e+06/ 3.2e+05
Total		1.3e+09/5.3e+09	1.8e+09/9.5e+09	1.1e+07/4.8e+07	1.7e+05/1.1e+06	5.1e+05/2.2e+05	7.8e+03/9.2e+04	1.8e+07/3.4e+06

Table 12: Per-pass performance for DecisionTree, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
classify Batch	F	5/7	29%	3.119 $\pm$ 0.0040	4.261 $\pm$ 0.0053	2.586$\pm$0.0043	20.86 $\pm$ 0.010	1.21$\times$	1.37$\times$	1.65$\times$	0.20 $\times$
classify Depth	F	4/7	43%	0.0448 $\pm$ 0.00002	0.0611 $\pm$ 0.00001	0.0391$\pm$0.00112	0.3004 $\pm$ 0.00193	1.15$\times$	1.36$\times$	1.56$\times$	0.20 $\times$
classify tree	F	5/7	29%	0.0447 $\pm$ 0.00001	0.0609 $\pm$ 0.00013	0.0391$\pm$0.00006	0.2976 $\pm$ 0.00114	1.14$\times$	1.36$\times$	1.56$\times$	0.20 $\times$
countFeatureUses	F	3/7	57%	0.0479 $\pm$ 0.00002	0.1405 $\pm$ 0.00120	0.0317$\pm$0.00003	0.3785 $\pm$ 0.00208	1.51$\times$	2.93$\times$	4.44$\times$	0.37 $\times$
countLeaves	F	2/7	71%	0.0455 $\pm$ 0.00002	0.1385 $\pm$ 0.00020	0.0249$\pm$0.00002	0.3693 $\pm$ 0.00116	1.82$\times$	3.04$\times$	5.55$\times$	0.38 $\times$
countNodes	F	2/7	71%	0.0450 $\pm$ 0.00143	0.1388 $\pm$ 0.00010	0.0260$\pm$0.00003	0.3691 $\pm$ 0.00119	1.73$\times$	3.09$\times$	5.33$\times$	0.38 $\times$
countSmallLeaves	F	3/7	57%	0.0485 $\pm$ 0.00001	0.1347 $\pm$ 0.00171	0.0322$\pm$0.00001	0.3718 $\pm$ 0.00164	1.50$\times$	2.78$\times$	4.18$\times$	0.36 $\times$
inferenceCost	F	2/7	71%	0.0485 $\pm$ 0.00088	0.1371 $\pm$ 0.00167	0.0290$\pm$0.00001	0.3701 $\pm$ 0.00118	1.67$\times$	2.83$\times$	4.73$\times$	0.37 $\times$
sumImpurity	F	3/7	57%	0.0480 $\pm$ 0.00001	0.1375 $\pm$ 0.00003	0.0323$\pm$0.00100	0.3811 $\pm$ 0.00256	1.48$\times$	2.86$\times$	4.25$\times$	0.36 $\times$
sumPathLengths	F	3/7	57%	0.0505 $\pm$ 0.00002	0.1373 $\pm$ 0.00007	0.0336$\pm$0.00002	0.3874 $\pm$ 0.00195	1.50$\times$	2.72$\times$	4.09$\times$	0.35 $\times$
sumSamples	F	3/7	57%	0.0459 $\pm$ 0.00001	0.1333 $\pm$ 0.00019	0.0318$\pm$0.00002	0.3844 $\pm$ 0.00025	1.44$\times$	2.91$\times$	4.20$\times$	0.35 $\times$
treeDepth	F	2/7	71%	0.0487 $\pm$ 0.00069	0.1370 $\pm$ 0.00120	0.0279$\pm$0.00001	0.3701 $\pm$ 0.00118	1.74$\times$	2.81$\times$	4.90$\times$	0.37 $\times$
Total				3.637	5.618	2.934	24.84	1.24$\times$	1.54$\times$	1.91$\times$	0.23 $\times$
Geomean								1.48$\times$	2.39$\times$	3.53$\times$	0.32 $\times$

Table 13: Per-pass PAPI counters for DecisionTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	1.6e+10/ 1.3e+10	3.3e+10 /5.9e+10	4.6e+08/ 2.7e+07	1.0e+05/ 4.1e+04	1.7e+06 /2.0e+06	1.0e+05/ 3.8e+04	9.3e+08/ 1.8e+07
classify Depth	F	2.3e+08/ 2.0e+08	4.7e+08 /8.5e+08	6.6e+06/ 4.0e+05	8.2e+02/ 5.7e+02	1.9e+04 /2.8e+04	8.3e+02/ 5.2e+02	1.3e+07/ 2.8e+05
classify tree	F	2.3e+08/ 2.0e+08	4.7e+08 /8.5e+08	6.6e+06/ 4.0e+05	8.2e+02/ 4.0e+02	2.2e+04 /2.8e+04	8.2e+02/ 3.6e+02	1.3e+07/ 2.9e+05
countFeatureUses	F	2.4e+08/ 1.6e+08	7.5e+08 /7.8e+08	7.6e+06/ 3.6e+05	8.4e+02/ 7.5e+02	3.9e+04/ 1.4e+04	8.4e+02/ 7.5e+02	1.5e+07/ 2.0e+06
countLeaves	F	2.3e+08/ 1.3e+08	6.0e+08 /6.1e+08	7.4e+06/ 1.0e+05	9.3e+02/ 3.2e+02	4.6e+04/ 7.8e+03	9.3e+02/ 3.2e+02	1.4e+07/ 3.3e+05
countNodes	F	2.3e+08/ 1.3e+08	6.1e+08 /6.2e+08	7.3e+06/ 1.1e+05	7.2e+02/ 3.5e+02	4.8e+04/ 7.9e+03	7.2e+02/ 3.4e+02	1.4e+07/ 3.5e+05
countSmallLeaves	F	2.5e+08/ 1.6e+08	7.1e+08 /8.4e+08	5.8e+06/ 8.8e+04	1.0e+03/ 8.5e+02	3.1e+04/ 1.3e+04	1.0e+03/ 8.5e+02	1.5e+07/ 2.0e+06
inferenceCost	F	2.5e+08/ 1.5e+08	6.1e+08/ 6.1e+08	8.0e+06/ 1.2e+05	1.1e+03/ 4.3e+02	2.0e+04/ 7.7e+03	1.1e+03/ 4.2e+02	1.4e+07/ 3.3e+05
sumImpurity	F	2.4e+08/ 1.6e+08	7.6e+08 /7.8e+08	7.8e+06/ 1.1e+06	1.3e+03/ 8.2e+02	3.8e+04/ 1.3e+04	1.3e+03/ 8.2e+02	1.4e+07/ 2.1e+06
sumPathLengths	F	2.6e+08/ 1.7e+08	8.0e+08 /9.0e+08	7.5e+06/ 4.7e+05	9.8e+02/ 7.1e+02	3.6e+04/ 1.4e+04	9.7e+02/ 7.1e+02	1.5e+07/ 2.0e+06
sumSamples	F	2.3e+08/ 1.6e+08	6.4e+08 /7.7e+08	6.2e+06/ 2.5e+05	1.0e+03/ 7.1e+02	4.0e+04/ 1.4e+04	1.0e+03/ 7.2e+02	1.5e+07/ 2.0e+06
treeDepth	F	2.5e+08/ 1.4e+08	6.1e+08/ 6.1e+08	8.1e+06/ 1.1e+05	1.0e+03/ 4.6e+02	1.6e+04/ 7.8e+03	1.0e+03/ 4.4e+02	1.4e+07/ 3.3e+05
Total		1.8e+10/1.5e+10	4.0e+10/6.8e+10	5.4e+08/3.0e+07	1.1e+05/4.7e+04	2.0e+06/2.1e+06	1.1e+05/4.4e+04	1.1e+09/2.9e+07

Table 14: Per-pass PAPI counters for DecisionTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
classify Batch	F	2.2e+10 /1.1e+11	5.1e+10 /1.9e+11	1.7e+07 /4.7e+07	2.2e+04 /1.3e+05	3.5e+05 /2.2e+06	2.1e+04/ 1.8e+04	8.5e+08/ 1.9e+07
classify Depth	F	3.1e+08 /1.5e+09	7.4e+08 /2.7e+09	2.5e+05 /6.4e+05	3.2e+02 /3.3e+03	5.1e+03 /3.0e+04	3.2e+02 /6.5e+02	1.2e+07/ 3.2e+05
classify tree	F	3.1e+08 /1.5e+09	7.4e+08 /2.7e+09	2.5e+05 /6.8e+05	3.0e+02 /1.7e+03	5.0e+03 /3.2e+04	3.0e+02/ 1.7e+02	1.2e+07/ 3.0e+05
countFeatureUses	F	7.1e+08 /1.9e+09	1.3e+09 /3.1e+09	2.0e+06 /3.2e+06	5.8e+02 /2.0e+03	5.9e+03 /6.9e+04	5.7e+02 /8.5e+02	9.3e+06/ 1.2e+06
countLeaves	F	7.0e+08 /1.9e+09	1.2e+09 /2.9e+09	4.4e+06/ 6.7e+05	2.9e+03/ 2.7e+03	9.5e+03 /3.2e+04	2.9e+03/ 2.4e+02	9.2e+06/ 3.3e+05
countNodes	F	7.0e+08 /1.9e+09	1.2e+09 /3.0e+09	4.4e+06/ 6.8e+05	8.3e+02 /2.0e+03	5.4e+03 /3.2e+04	8.3e+02/ 3.8e+02	9.2e+06/ 3.3e+05
countSmallLeaves	F	6.8e+08 /1.9e+09	1.3e+09 /2.9e+09	1.1e+06 /3.4e+06	1.1e+03 /1.9e+03	4.9e+03 /7.3e+04	1.1e+03/ 7.0e+02	9.3e+06/ 1.2e+06
inferenceCost	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.1e+05	4.4e+02 /1.5e+03	4.7e+03 /3.0e+04	4.4e+02/ 2.5e+02	9.2e+06/ 3.5e+05
sumImpurity	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	2.2e+06 /3.5e+06	2.5e+02 /3.4e+02	6.1e+03 /7.5e+04	2.5e+02 /3.1e+02	9.4e+06/ 1.2e+06
sumPathLengths	F	7.0e+08 /2.0e+09	1.3e+09 /3.1e+09	1.5e+06 /3.6e+06	4.9e+02 /3.2e+03	4.3e+03 /7.9e+04	4.9e+02 /9.9e+02	9.2e+06/ 1.2e+06
sumSamples	F	6.8e+08 /2.0e+09	1.2e+09 /3.0e+09	1.1e+06 /3.6e+06	6.9e+02 /2.4e+03	4.3e+03 /7.7e+04	7.0e+02 /8.1e+02	9.3e+06/ 1.2e+06
treeDepth	F	6.9e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.2e+05	2.3e+02 /1.5e+03	4.8e+03 /3.1e+04	2.3e+02/ 2.2e+02	9.2e+06/ 3.7e+05
Total		2.8e+10/1.3e+11	6.4e+10/2.2e+11	4.3e+07/6.9e+07	3.0e+04/1.5e+05	4.1e+05/2.8e+06	3.0e+04/2.4e+04	9.6e+08/2.7e+07

Table 15: Per-pass performance for DomTree, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
SumArea	F	6/15	60%	0.0350 $\pm$ 0.00002	0.0685 $\pm$ 0.00017	0.0191$\pm$0.00003	0.0893 $\pm$ 0.00016	1.83$\times$	1.96$\times$	3.59$\times$	0.77 $\times$
computeWidths	M	14/15	7%	0.0298$\pm$0.00004	0.0303 $\pm$ 0.00004	0.0441 $\pm$ 0.00005	0.0878 $\pm$ 0.00115	0.67 $\times$	1.02 $\times$	0.69 $\times$	0.34 $\times$
count styled	F	3/15	80%	0.0338 $\pm$ 0.00062	0.0682 $\pm$ 0.00002	0.0116$\pm$0.00001	0.0882 $\pm$ 0.00020	2.92$\times$	2.02$\times$	5.89$\times$	0.77 $\times$
find max Bottom	F	5/15	67%	0.0338 $\pm$ 0.00001	0.0679 $\pm$ 0.00097	0.0183$\pm$0.00004	0.0877 $\pm$ 0.00003	1.85$\times$	2.01$\times$	3.71$\times$	0.77 $\times$
scaleLayout	M	15/15	0%	0.0285$\pm$0.00004	0.0301 $\pm$ 0.00088	0.0345 $\pm$ 0.00004	0.0840 $\pm$ 0.00007	0.83 $\times$	1.05 $\times$	0.87 $\times$	0.36 $\times$
sumTextWidth	F	3/15	80%	0.0336 $\pm$ 0.00001	0.0688 $\pm$ 0.00002	0.0116$\pm$0.00002	0.0885 $\pm$ 0.00128	2.89$\times$	2.05$\times$	5.91$\times$	0.78 $\times$
Total				0.1945	0.3337	0.1392	0.5255	1.40$\times$	1.72$\times$	2.40$\times$	0.63 $\times$
Geomean								1.59$\times$	1.61$\times$	2.55$\times$	0.59 $\times$

Table 16: Per-pass PAPI counters for DomTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.8e+08/ 9.7e+07	2.9e+08 /4.9e+08	2.6e+06/ 5.1e+05	3.5e+02/ 1.2e+02	1.7e+05/ 1.2e+04	3.5e+02/ 1.2e+02	1.4e+07/ 1.9e+06
computeWidths	M	6.5e+07 /1.4e+08	1.4e+08 /4.6e+08	1.2e+06 /1.5e+06	5.1e+04 /6.1e+04	5.1e+03 /1.1e+04	1.4e+03 /1.5e+04	1.2e+06/ 4.3e+05
count styled	F	1.7e+08/ 5.9e+07	2.6e+08 /2.6e+08	2.0e+06/ 2.7e+04	5.5e+02/ 7.3e+01	2.8e+05/ 5.0e+03	5.5e+02/ 7.9e+01	1.4e+07/ 6.4e+05
find max Bottom	F	1.7e+08/ 9.2e+07	2.5e+08 /4.1e+08	2.3e+06/ 1.4e+06	4.8e+02/ 7.5e+01	1.6e+05/ 9.6e+03	4.9e+02/ 7.7e+01	1.4e+07/ 1.4e+06
scaleLayout	M	5.9e+07 /9.0e+07	8.8e+07 /2.9e+08	1.1e+06/ 7.4e+05	6.2e+04/ 4.6e+04	1.5e+04/ 8.8e+03	1.3e+03 /1.4e+04	1.3e+06/ 4.5e+05
sumTextWidth	F	1.7e+08/ 5.9e+07	2.0e+08 /2.4e+08	2.4e+06/ 1.9e+04	5.3e+02/ 7.6e+01	4.5e+05/ 4.2e+03	5.3e+02/ 7.8e+01	1.4e+07/ 6.5e+05
Total		8.1e+08/5.3e+08	1.2e+09/2.2e+09	1.2e+07/4.2e+06	1.1e+05/1.1e+05	1.1e+06/5.0e+04	4.6e+03/3.0e+04	5.7e+07/5.4e+06

Table 17: Per-pass PAPI counters for DomTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
SumArea	F	3.5e+08 /4.5e+08	4.4e+08 /1.8e+09	8.1e+06/ 2.2e+06	1.7e+02 /9.6e+02	1.3e+04 /3.2e+04	1.7e+02 /9.6e+02	1.0e+07/ 1.4e+06
computeWidths	M	6.7e+07 /3.6e+08	1.5e+08 /9.0e+08	1.4e+06 /3.5e+06	4.6e+04 /2.4e+05	3.9e+03 /1.6e+04	1.4e+03 /2.2e+04	1.1e+06/ 4.4e+05
count styled	F	3.5e+08 /4.5e+08	4.3e+08 /1.7e+09	1.0e+07/ 1.3e+06	1.7e+02 /4.1e+02	3.5e+04 /3.7e+04	1.6e+02 /3.4e+02	1.0e+07/ 4.2e+05
find max Bottom	F	3.4e+08 /4.4e+08	4.6e+08 /1.8e+09	1.0e+07/ 1.5e+06	9.2e+01 /6.2e+02	1.4e+04 /2.8e+04	9.2e+01 /6.0e+02	1.0e+07/ 9.5e+05
scaleLayout	M	6.7e+07 /3.4e+08	1.1e+08 /7.6e+08	9.7e+05 /2.9e+06	6.1e+04 /8.5e+04	4.4e+03 /1.2e+04	1.8e+03 /2.4e+04	1.1e+06/ 4.6e+05
sumTextWidth	F	3.5e+08 /4.5e+08	3.9e+08 /1.7e+09	1.1e+07/ 1.3e+06	6.2e+02 /9.4e+02	8.8e+04/ 3.9e+04	6.2e+02 /7.2e+02	1.0e+07/ 4.1e+05
Total		1.5e+09/2.5e+09	2.0e+09/8.6e+09	4.1e+07/1.3e+07	1.1e+05/3.3e+05	1.6e+05/1.7e+05	4.2e+03/4.9e+04	4.3e+07/4.0e+06

Table 18: Per-pass performance for KDTree, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Find nearest Neighbour	F	13/17	24%	0.0271 $\pm$ 0.00004	0.0391 $\pm$ 0.00012	0.0213$\pm$0.00001	0.1184 $\pm$ 0.00114	1.27$\times$	1.45$\times$	1.84$\times$	0.33 $\times$
countInRange tight_box	F	13/17	24%	0.0272 $\pm$ 0.00001	0.0397 $\pm$ 0.00091	0.0200$\pm$0.00007	0.1241 $\pm$ 0.00003	1.36$\times$	1.46$\times$	1.99$\times$	0.32 $\times$
photonMappingBenchmark	F	12/17	29%	0.0416 $\pm$ 0.00001	0.0473 $\pm$ 0.00001	0.0414$\pm$0.00002	0.1306 $\pm$ 0.00130	1.01 $\times$	1.14$\times$	1.14$\times$	0.36 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0243 $\pm$ 0.00002	0.0382 $\pm$ 0.00001	0.0165$\pm$0.00001	0.1142 $\pm$ 0.00105	1.47$\times$	1.58$\times$	2.31$\times$	0.33 $\times$
sumMassInRange	F	14/17	18%	0.0265 $\pm$ 0.00125	0.0397 $\pm$ 0.00090	0.0204$\pm$0.00009	0.1242 $\pm$ 0.00115	1.30$\times$	1.50$\times$	1.95$\times$	0.32 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0281 $\pm$ 0.00002	0.0388 $\pm$ 0.00001	0.0207$\pm$0.00001	0.1142 $\pm$ 0.00003	1.35$\times$	1.38$\times$	1.87$\times$	0.34 $\times$
Total				0.1746	0.2428	0.1402	0.7257	1.25$\times$	1.39$\times$	1.73$\times$	0.33 $\times$
Geomean								1.28$\times$	1.41$\times$	1.81$\times$	0.33 $\times$

Table 19: Per-pass PAPI counters for KDTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.4e+08/ 1.1e+08	4.7e+08 /6.3e+08	2.9e+06/ 1.4e+06	2.5e+02/ 1.5e+02	5.0e+03 /1.9e+04	2.5e+02/ 1.5e+02	7.7e+06/ 1.7e+06
countInRange tight_box	F	1.4e+08/ 1.0e+08	4.0e+08 /5.5e+08	2.2e+06/ 1.9e+06	1.2e+02/ 1.0e+02	3.6e+03 /2.1e+04	1.2e+02/ 1.0e+02	7.7e+06/ 1.7e+06
photonMappingBenchmark	F	2.1e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.1e+06 /2.5e+06	2.0e+02 /4.5e+02	3.4e+03 /1.8e+04	2.0e+02 /4.5e+02	5.6e+06/ 1.5e+06
pointCloudNeighborhood	F	1.2e+08/ 8.4e+07	3.3e+08 /4.8e+08	2.6e+06/ 8.0e+05	7.5e+01 /9.0e+01	4.0e+03 /2.0e+04	7.6e+01 /8.9e+01	7.8e+06/ 1.5e+06
sumMassInRange	F	1.3e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.6e+06/ 2.4e+06	7.3e+01 /1.3e+02	3.8e+03 /2.3e+04	7.3e+01 /1.3e+02	7.7e+06/ 1.9e+06
twoPointCorrelation bin_8_16	F	1.4e+08/ 1.1e+08	4.8e+08 /6.1e+08	2.2e+06/ 6.3e+05	9.5e+01 /1.1e+02	2.9e+03 /1.6e+04	9.7e+01 /1.1e+02	7.7e+06/ 1.3e+06
Total		8.8e+08/7.1e+08	3.1e+09/4.0e+09	1.5e+07/9.6e+06	8.2e+02/1.0e+03	2.3e+04/1.2e+05	8.2e+02/1.0e+03	4.4e+07/9.6e+06

Table 20: Per-pass PAPI counters for KDTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Find nearest Neighbour	F	2.0e+08 /6.0e+08	4.9e+08 /1.3e+09	4.7e+06 /5.2e+06	1.4e+02 /1.5e+03	5.1e+03 /2.2e+04	1.4e+02 /1.5e+03	5.8e+06/ 1.5e+06
countInRange tight_box	F	2.0e+08 /6.3e+08	4.5e+08 /1.3e+09	5.1e+06 /5.8e+06	1.4e+02 /2.4e+03	5.3e+03 /2.2e+04	1.4e+02 /2.4e+03	5.6e+06/ 1.6e+06
photonMappingBenchmark	F	2.4e+08 /6.6e+08	1.1e+09 /2.0e+09	2.7e+06 /6.0e+06	4.4e+02 /1.9e+03	5.9e+03 /2.3e+04	4.4e+02 /1.8e+03	5.1e+06/ 1.5e+06
pointCloudNeighborhood	F	1.9e+08 /5.8e+08	4.1e+08 /1.3e+09	5.1e+06/ 4.7e+06	1.7e+02 /1.4e+03	5.6e+03 /1.9e+04	1.8e+02 /1.4e+03	5.8e+06/ 1.4e+06
sumMassInRange	F	2.0e+08 /6.3e+08	4.7e+08 /1.3e+09	5.1e+06 /6.3e+06	1.8e+02 /1.7e+03	4.8e+03 /2.6e+04	1.9e+02 /1.7e+03	5.6e+06/ 1.6e+06
twoPointCorrelation bin_8_16	F	2.0e+08 /5.8e+08	6.1e+08 /1.4e+09	4.8e+06/ 3.7e+06	2.2e+02 /1.4e+03	6.5e+03 /1.8e+04	2.3e+02 /1.4e+03	5.8e+06/ 1.3e+06
Total		1.2e+09/3.7e+09	3.5e+09/8.4e+09	2.8e+07/3.2e+07	1.3e+03/1.0e+04	3.3e+04/1.3e+05	1.3e+03/1.0e+04	3.4e+07/8.8e+06

Table 21: Per-pass performance for LinearListReduction, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
reduction	F	2/11	82%	0.0292 $\pm$ 0.00002	0.1382 $\pm$ 0.00130	0.0050$\pm$0.00001	<i>OOM</i>	5.83$\times$	4.74$\times$	27.62$\times$	–
Total				0.0292	0.1382	0.0050	–	5.83$\times$	4.74$\times$	27.62$\times$	–
Geomean								5.83$\times$	4.74$\times$	27.62$\times$	–

Table 22: Per-pass PAPI counters for LinearListReduction (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.5e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.5e+06/ 4.7e+04	3.6e+02/ 4.7e+01	2.8e+05/ 6.9e+03	3.6e+02/ 4.7e+01	1.2e+07/ 1.1e+06
Total		1.5e+08/2.5e+07	9.0e+07/1.0e+08	1.5e+06/4.7e+04	3.6e+02/4.7e+01	2.8e+05/6.9e+03	3.6e+02/4.7e+01	1.2e+07/1.1e+06

Table 23: Per-pass PAPI counters for LinearListReduction (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
reduction	F	7.0e+08/–	2.5e+08/–	5.0e+06/–	3.9e+02/–	5.4e+05/–	3.9e+02/–	2.6e+07/–
Total		7.0e+08/–	2.5e+08/–	5.0e+06/–	3.9e+02/–	5.4e+05/–	3.9e+02/–	2.6e+07/–

Table 24: Per-pass performance for `List`, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1 List	M	2/2	0%	0.2562$\pm$0.00250	<i>OOM</i>	0.3404 $\pm$ 0.00459	<i>OOM</i>	0.75 $\times$	–	–	–
length List	M	1/2	50%	0.0410 $\pm$ 0.00076	<i>OOM</i>	0.0215$\pm$0.00001	<i>OOM</i>	1.90$\times$	–	–	–
sumList	F	2/2	0%	0.0440$\pm$0.00001	<i>OOM</i>	0.0450 $\pm$ 0.00040	<i>OOM</i>	0.98 $\times$	–	–	–
sumList tail recursive	F	2/2	0%	0.0442$\pm$0.00002	<i>OOM</i>	0.0445 $\pm$ 0.00001	<i>OOM</i>	0.99 $\times$	–	–	–
Total				0.3853	–	0.4514	–	0.85 $\times$	–	–	–
Geomean								1.09 $\times$	–	–	–

Table 25: Per-pass PAPI counters for `List` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	5.9e+08 /9.8e+08	1.9e+09 /2.8e+09	8.0e+06/ 1.7e+06	4.1e+05 /5.3e+05	1.4e+04 /5.6e+04	6.4e+03 /9.8e+03	9.2e+06/ 7.8e+06
length List	M	2.1e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.2e+05/ 8.4e+04	1.5e+02/ 6.5e+01	2.8e+03 /1.4e+04	1.6e+02/ 6.5e+01	1.3e+07/ 1.1e+06
sumList	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	6.9e+05/ 5.4e+05	8.9e+01/ 6.9e+01	3.1e+03 /6.9e+04	9.0e+01/ 7.0e+01	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	5.8e+05/ 5.6e+05	1.2e+02 /1.5e+02	3.0e+03 /5.9e+04	1.2e+02 /1.5e+02	1.3e+07/ 1.2e+07
Total		1.2e+09/1.5e+09	4.5e+09/5.6e+09	9.4e+06/2.8e+06	4.1e+05/5.4e+05	2.3e+04/2.0e+05	6.7e+03/1.0e+04	4.8e+07/3.3e+07

Table 26: Per-pass PAPI counters for `List` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Total		–/–	–/–	–/–	–/–	–/–	–/–	–/–

Table 27: Per-pass performance for `MonoTree`, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1Tree	M	3/3	0%	0.0455 $\pm$ 0.00039	0.0604 $\pm$ 0.00042	0.0428$\pm$0.00040	0.1126 $\pm$ 0.00141	1.06 $\times$	1.33$\times$	1.41$\times$	0.54 $\times$
sumTree	F	3/3	0%	0.0088$\pm$0.00001	0.0405 $\pm$ 0.00001	0.0114 $\pm$ 0.00001	0.0806 $\pm$ 0.00115	0.78 $\times$	4.59$\times$	3.56$\times$	0.50 $\times$
sumTree TailRec	F	3/3	0%	0.0083$\pm$0.00000	0.0402 $\pm$ 0.00002	0.0111 $\pm$ 0.00002	0.0806 $\pm$ 0.00002	0.75 $\times$	4.86$\times$	3.63$\times$	0.50 $\times$
Total				0.0626	0.1412	0.0652	0.2738	0.96 $\times$	2.26$\times$	2.17$\times$	0.52 $\times$
Geomean								0.85 $\times$	3.09$\times$	2.64$\times$	0.51 $\times$

Table 28: Per-pass PAPI counters for `MonoTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.6e+08/ 1.5e+08	5.0e+08 /5.9e+08	2.5e+05 /4.1e+05	3.9e+04 /4.6e+04	1.7e+03 /5.1e+03	1.1e+03 /1.8e+03	5.1e+05/ 4.5e+05
sumTree	F	4.5e+07 /5.8e+07	2.2e+08 /2.6e+08	6.3e+03 /3.6e+04	1.7e+02 /1.9e+02	7.0e+02 /4.0e+03	1.7e+02 /2.0e+02	7.7e+05/ 6.2e+05
sumTree TailRec	F	4.2e+07 /5.6e+07	1.9e+08 /2.2e+08	6.0e+03 /2.4e+04	2.4e+02/ 2.2e+02	7.9e+02 /4.0e+03	2.4e+02/ 2.2e+02	7.7e+05/ 6.1e+05
Total		2.5e+08/2.6e+08	9.1e+08/1.1e+09	2.6e+05/4.7e+05	4.0e+04/4.6e+04	3.2e+03/1.3e+04	1.5e+03/2.2e+03	2.0e+06/1.7e+06

Table 29: Per-pass PAPI counters for `MonoTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add1Tree	M	2.4e+08 /5.0e+08	7.1e+08 /1.2e+09	1.5e+05 /9.8e+05	3.1e+04 /5.0e+04	1.9e+03 /1.6e+04	1.6e+03 /2.1e+03	4.1e+05/ 3.9e+05
sumTree	F	2.0e+08 /4.1e+08	4.3e+08 /6.3e+08	9.0e+04 /8.8e+05	1.8e+02 /2.2e+02	1.1e+03 /1.3e+04	1.8e+02 /2.0e+02	3.3e+05 /3.6e+05
sumTree TailRec	F	2.0e+08 /4.1e+08	4.1e+08 /6.4e+08	6.8e+04 /8.7e+05	3.5e+02/ 1.1e+02	1.1e+03 /1.3e+04	3.5e+02/ 1.1e+02	3.2e+05 /3.6e+05
Total		6.5e+08/1.3e+09	1.6e+09/2.5e+09	3.1e+05/2.7e+06	3.2e+04/5.0e+04	4.2e+03/4.2e+04	2.1e+03/2.4e+03	1.1e+06/1.1e+06

Table 30: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
countLargeItems	F	3/5	40%	0.0125 $\pm$ 0.00001	0.0442 $\pm$ 0.00001	0.0104$\pm$0.00069	0.0464 $\pm$ 0.00001	1.20$\times$	3.55$\times$	4.24$\times$	0.95 $\times$
countMarked	F	3/5	40%	0.0125 $\pm$ 0.00000	0.0442 $\pm$ 0.00001	0.0107$\pm$0.00075	0.0467 $\pm$ 0.00096	1.16$\times$	3.55$\times$	4.12$\times$	0.95 $\times$
countSurvivors	F	4/5	20%	0.0135 $\pm$ 0.00067	0.0444 $\pm$ 0.00001	0.0129$\pm$0.00001	0.0468 $\pm$ 0.00001	1.04 $\times$	3.30$\times$	3.44$\times$	0.95 $\times$
deadBytes	F	4/5	20%	0.0131 $\pm$ 0.00001	0.0436 $\pm$ 0.00084	0.0124$\pm$0.00001	0.0470 $\pm$ 0.00002	1.06 $\times$	3.33$\times$	3.52$\times$	0.93 $\times$
liveBytes	F	4/5	20%	0.0131 $\pm$ 0.00060	0.0430 $\pm$ 0.00001	0.0124$\pm$0.00001	0.0470 $\pm$ 0.00001	1.05 $\times$	3.29$\times$	3.46$\times$	0.91 $\times$
sumObjIds	F	3/5	40%	0.0125 $\pm$ 0.00001	0.0428 $\pm$ 0.00001	0.0104$\pm$0.00002	0.0471 $\pm$ 0.00103	1.20$\times$	3.44$\times$	4.12$\times$	0.91 $\times$
sweepUnmarked	M	5/5	0%	0.0722$\pm$0.00007	0.0872 $\pm$ 0.00006	0.0808 $\pm$ 0.00017	0.1858 $\pm$ 0.00127	0.89 $\times$	1.21$\times$	1.08 $\times$	0.47 $\times$
totalHeapSize	F	3/5	40%	0.0123 $\pm$ 0.00026	0.0428 $\pm$ 0.00005	0.0105$\pm$0.00097	0.0472 $\pm$ 0.00092	1.17$\times$	3.47$\times$	4.07$\times$	0.91 $\times$
touchHotObjects	M	5/5	0%	0.0714$\pm$0.00089	0.0868 $\pm$ 0.00095	0.0804 $\pm$ 0.00002	0.1899 $\pm$ 0.00181	0.89 $\times$	1.22$\times$	1.08 $\times$	0.46 $\times$
Total				0.2330	0.4790	0.2410	0.7039	0.97 $\times$	2.06$\times$	1.99$\times$	0.68 $\times$
Geomean								1.07 $\times$	2.71$\times$	2.89$\times$	0.80 $\times$

Table 31: Per-pass PAPI counters for `ObjectGraph` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	6.3e+07/ 5.3e+07	2.6e+08 /2.8e+08	1.2e+06/ 6.7e+04	2.2e+02 /2.4e+02	3.3e+03 /3.6e+03	2.1e+02 /2.4e+02	2.9e+06/ 6.4e+05
countMarked	F	6.3e+07/ 5.4e+07	2.6e+08 /2.8e+08	1.3e+06/ 5.3e+04	1.7e+02 /2.1e+02	3.3e+03 /3.8e+03	1.7e+02 /2.1e+02	2.9e+06/ 6.3e+05
countSurvivors	F	6.8e+07/ 6.6e+07	3.1e+08 /3.8e+08	9.8e+05/ 7.3e+04	3.5e+02/ 1.7e+02	2.5e+03 /5.1e+03	3.5e+02/ 1.7e+02	2.7e+06/ 1.2e+06
deadBytes	F	6.6e+07/ 6.3e+07	2.7e+08 /3.5e+08	7.6e+05/ 5.7e+04	2.2e+02/ 1.5e+02	2.5e+03 /5.2e+03	2.2e+02/ 1.6e+02	2.7e+06/ 1.2e+06
liveBytes	F	6.6e+07/ 6.3e+07	2.7e+08 /3.5e+08	9.0e+05/ 5.5e+04	1.6e+02/ 1.5e+02	2.4e+03 /5.3e+03	1.6e+02/ 1.6e+02	2.7e+06/ 1.2e+06
sumObjIds	F	6.3e+07/ 5.3e+07	2.5e+08 /2.7e+08	1.2e+06/ 8.5e+04	3.2e+02/ 1.2e+02	2.8e+03 /4.2e+03	3.2e+02/ 1.2e+02	2.7e+06/ 6.1e+05
sweepUnmarked	M	1.9e+08 /2.3e+08	5.5e+08 /8.8e+08	9.6e+05/ 4.9e+05	8.6e+04 /1.1e+05	4.7e+03 /1.1e+04	5.0e+03 /7.4e+03	2.0e+06/ 1.0e+06
totalHeapSize	F	6.2e+07/ 5.3e+07	2.5e+08 /2.7e+08	1.3e+06/ 9.7e+04	1.6e+02/ 1.5e+02	2.5e+03 /3.5e+03	1.7e+02/ 1.5e+02	2.9e+06/ 6.4e+05
touchHotObjects	M	1.9e+08 /2.3e+08	5.7e+08 /9.0e+08	9.5e+05/ 4.8e+05	1.0e+05 /1.3e+05	4.8e+03 /1.1e+04	4.5e+03 /7.0e+03	2.0e+06/ 1.0e+06
Total		8.4e+08/8.7e+08	3.0e+09/4.0e+09	9.5e+06/1.5e+06	1.9e+05/2.4e+05	2.9e+04/5.3e+04	1.1e+04/1.6e+04	2.4e+07/8.2e+06

Table 32: Per-pass PAPI counters for `ObjectGraph` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
countLargeItems	F	2.2e+08 /2.4e+08	4.6e+08 /7.9e+08	5.6e+05/ 4.8e+05	2.5e+02/ 6.1e+01	4.6e+03 /7.9e+03	2.5e+02/ 6.4e+01	1.8e+06/ 3.7e+05
countMarked	F	2.2e+08 /2.4e+08	4.6e+08 /7.9e+08	5.7e+05/ 4.9e+05	1.0e+02 /1.3e+02	4.8e+03 /9.5e+03	1.0e+02 /1.3e+02	1.7e+06/ 3.8e+05
countSurvivors	F	2.3e+08 /2.4e+08	5.5e+08 /8.2e+08	1.8e+05 /4.2e+05	1.8e+02 /2.3e+02	1.9e+03 /6.6e+03	1.8e+02 /2.3e+02	1.8e+06/ 6.9e+05
deadBytes	F	2.2e+08 /2.4e+08	4.9e+08 /8.1e+08	3.9e+05 /4.4e+05	1.5e+02 /3.1e+02	3.7e+03 /6.4e+03	1.5e+02 /3.1e+02	1.7e+06/ 6.7e+05
liveBytes	F	2.2e+08 /2.4e+08	5.0e+08 /8.1e+08	1.6e+05 /4.3e+05	1.5e+02 /3.2e+02	2.2e+03 /6.3e+03	1.5e+02 /3.2e+02	1.8e+06/ 6.8e+05
sumObjIds	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.6e+05/ 5.0e+05	2.4e+02/ 9.9e+01	4.7e+03 /8.9e+03	2.5e+02/ 9.9e+01	1.7e+06/ 3.6e+05
sweepUnmarked	M	2.7e+08 /7.7e+08	7.6e+08 /2.1e+09	1.9e+05 /2.5e+06	7.6e+04 /1.5e+05	3.3e+03 /2.2e+04	2.4e+03 /5.2e+03	1.8e+06/ 9.3e+05
totalHeapSize	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.5e+05/ 5.5e+05	9.9e+01 /3.7e+02	4.6e+03 /9.2e+03	9.9e+01 /3.6e+02	1.8e+06/ 3.7e+05
touchHotObjects	M	2.7e+08 /7.9e+08	7.7e+08 /2.2e+09	2.1e+05 /2.0e+06	1.0e+05 /1.1e+05	3.2e+03 /2.0e+04	5.0e+03 /5.3e+03	1.8e+06/ 9.7e+05
Total		2.1e+09/3.2e+09	4.9e+09/9.9e+09	3.4e+06/7.8e+06	1.8e+05/2.6e+05	3.3e+04/9.7e+04	8.6e+03/1.2e+04	1.6e+07/5.4e+06

Table 33: Per-pass performance for `OctTree`, ADT has 16 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
barnesHutPotential	F	11/16	31%	0.0052 $\pm$ 0.00000	0.0073 $\pm$ 0.00000	0.0049$\pm$0.00000	<i>OOM</i>	1.06 $\times$	1.41$\times$	1.50$\times$	–
clearFlags	M	15/16	6%	0.0196 $\pm$ 0.00002	0.0186$\pm$0.00003	0.0219 $\pm$ 0.00002	<i>OOM</i>	0.89 $\times$	0.95 $\times$	0.85 $\times$	–
countActive	F	10/16	38%	0.0043 $\pm$ 0.00000	0.0057 $\pm$ 0.00000	0.0036$\pm$0.00000	<i>OOM</i>	1.21$\times$	1.32$\times$	1.59$\times$	–
countParticles	F	8/16	50%	0.0043 $\pm$ 0.00000	0.0055 $\pm$ 0.00000	0.0033$\pm$0.00000	<i>OOM</i>	1.30$\times$	1.29$\times$	1.67$\times$	–
fmmPotential	F	12/16	25%	0.0121 $\pm$ 0.00001	0.0121 $\pm$ 0.00000	0.0119$\pm$0.00000	<i>OOM</i>	1.01 $\times$	1.00 $\times$	1.02 $\times$	–
scaleEnergy	M	16/16	0%	0.0192 $\pm$ 0.00002	0.0190$\pm$0.00002	0.0220 $\pm$ 0.00002	<i>OOM</i>	0.87 $\times$	0.99 $\times$	0.86 $\times$	–
sumEnergy	F	12/16	25%	0.0045 $\pm$ 0.00000	0.0067 $\pm$ 0.00000	0.0040$\pm$0.00000	<i>OOM</i>	1.12$\times$	1.51$\times$	1.70$\times$	–
sumMass	F	10/16	38%	0.0042 $\pm$ 0.00023	0.0055 $\pm$ 0.00035	0.0033$\pm$0.00005	<i>OOM</i>	1.28$\times$	1.30$\times$	1.67$\times$	–
Total				0.0733	0.0804	0.0749	–	0.98 $\times$	1.10 $\times$	1.07 $\times$	–
Geomean								1.08 $\times$	1.21$\times$	1.30$\times$	–

Table 34: Per-pass PAPI counters for `OctTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
barnesHutPotential	F	2.6e+07/ 2.5e+07	1.0e+08 /1.3e+08	8.5e+03 /5.3e+04	7.7e+01/ 4.1e+01	2.4e+02 /2.9e+03	8.0e+01/ 4.3e+01	5.7e+05/ 1.7e+05
clearFlags	M	4.6e+07 /5.5e+07	1.0e+08 /2.2e+08	1.1e+05 /2.8e+05	2.5e+04 /5.0e+04	1.2e+03 /4.3e+03	4.0e+02 /1.2e+03	5.8e+05/ 2.6e+05
countActive	F	2.2e+07/ 1.8e+07	7.3e+07/ 7.2e+07	1.1e+05/ 4.7e+04	8.9e+01/ 1.8e+01	5.2e+02 /2.0e+03	8.9e+01/ 1.8e+01	5.8e+05/ 8.6e+02
countParticles	F	2.2e+07/ 1.7e+07	7.1e+07/ 7.1e+07	1.2e+05/ 2.5e+04	7.7e+01/ 1.0e+01	5.5e+02 /8.3e+02	8.0e+01/ 1.0e+01	5.7e+05/ 8.5e+01
fmmPotential	F	6.1e+07/ 6.1e+07	2.0e+08 /2.3e+08	3.4e+03 /5.5e+04	1.0e+02 /1.1e+02	2.6e+02 /3.2e+03	1.0e+02 /1.1e+02	4.4e+05/ 2.1e+05
scaleEnergy	M	4.4e+07 /5.5e+07	1.1e+08 /2.3e+08	1.7e+05 /3.3e+05	3.5e+04 /6.0e+04	1.3e+03 /4.7e+03	5.2e+02 /1.2e+03	5.8e+05/ 2.7e+05
sumEnergy	F	2.2e+07/ 2.0e+07	9.6e+07 /1.1e+08	4.3e+04 /4.9e+04	5.4e+01/ 5.1e+01	2.8e+02 /3.3e+03	5.5e+01/ 5.2e+01	6.0e+05/ 1.9e+05
sumMass	F	2.1e+07/ 1.7e+07	7.1e+07 /7.9e+07	9.7e+04/ 2.6e+04	6.0e+01/ 4.1e+01	3.6e+02 /1.2e+03	6.0e+01/ 4.1e+01	6.1e+05/ 1.9e+04
Total		2.6e+08/2.7e+08	8.2e+08/1.1e+09	6.6e+05/8.6e+05	6.1e+04/1.1e+05	4.7e+03/2.2e+04	1.4e+03/2.6e+03	4.5e+06/1.1e+06

Table 35: Per-pass PAPI counters for `OctTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
barnesHutPotential	F	3.7e+07/–	1.1e+08/–	8.0e+03/–	5.8e+01/–	2.9e+02/–	5.8e+01/–	5.0e+05/–
clearFlags	M	4.3e+07/–	1.0e+08/–	8.0e+04/–	2.6e+04/–	1.0e+03/–	7.2e+02/–	5.8e+05/–
countActive	F	2.9e+07/–	8.8e+07/–	3.5e+04/–	4.7e+01/–	3.0e+02/–	4.7e+01/–	5.4e+05/–
countParticles	F	2.8e+07/–	8.1e+07/–	5.0e+04/–	4.2e+01/–	3.3e+02/–	4.2e+01/–	5.5e+05/–
fmmPotential	F	6.1e+07/–	2.1e+08/–	5.6e+03/–	5.7e+01/–	2.5e+02/–	5.7e+01/–	4.4e+05/–
scaleEnergy	M	4.5e+07/–	1.2e+08/–	9.6e+04/–	2.5e+04/–	1.0e+03/–	5.2e+02/–	5.7e+05/–
sumEnergy	F	3.4e+07/–	1.1e+08/–	9.3e+03/–	5.9e+01/–	3.1e+02/–	5.9e+01/–	5.3e+05/–
sumMass	F	2.8e+07/–	8.3e+07/–	6.2e+04/–	5.3e+01/–	3.0e+02/–	5.6e+01/–	5.6e+05/–
Total		3.0e+08/–	9.1e+08/–	3.4e+05/–	5.2e+04/–	3.8e+03/–	1.6e+03/–	4.3e+06/–

Table 36: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
addConstPW	M	8/8	0%	0.1175$\pm$0.00115	0.1295 $\pm$ 0.00271	0.1291 $\pm$ 0.00126	0.3972 $\pm$ 0.00168	0.91 $\times$	1.10$\times$	1.00 $\times$	0.33 $\times$
autorefineMaxLevel	F	4/8	50%	0.0200 $\pm$ 0.00001	0.0448 $\pm$ 0.00001	0.0195$\pm$0.00004	0.2334 $\pm$ 0.00129	1.03 $\times$	2.24$\times$	2.30$\times$	0.19 $\times$
compressMass	F	3/8	62%	0.0191 $\pm$ 0.00001	0.0452 $\pm$ 0.00002	0.0116$\pm$0.00001	0.2323 $\pm$ 0.00118	1.65$\times$	2.37$\times$	3.90$\times$	0.19 $\times$
diffPW	M	8/8	0%	0.1202$\pm$0.00016	0.1314 $\pm$ 0.00233	0.1296 $\pm$ 0.00173	0.3998 $\pm$ 0.00350	0.93 $\times$	1.09 $\times$	1.01 $\times$	0.33 $\times$
lbDeuxLoadProxy	F	5/8	38%	0.0209 $\pm$ 0.00001	0.0468 $\pm$ 0.00002	0.0185$\pm$0.00093	0.2372 $\pm$ 0.00123	1.13$\times$	2.24$\times$	2.53$\times$	0.20 $\times$
norm2Estimate	F	5/8	38%	0.0243 $\pm$ 0.00007	0.0544 $\pm$ 0.00006	0.0177$\pm$0.00092	0.1956 $\pm$ 0.00173	1.38$\times$	2.23$\times$	3.07$\times$	0.28 $\times$
pmapCutHistogram	F	4/8	50%	0.0313 $\pm$ 0.00001	0.0551 $\pm$ 0.00018	0.0125$\pm$0.00066	0.2408 $\pm$ 0.00118	2.49$\times$	1.76$\times$	4.39$\times$	0.23 $\times$
truncateTolViolations	F	3/8	62%	0.0193 $\pm$ 0.00001	0.0458 $\pm$ 0.00001	0.0115$\pm$0.00073	0.1998 $\pm$ 0.00018	1.67$\times$	2.37$\times$	3.97$\times$	0.23 $\times$
Total				0.3726	0.5529	0.3499	2.136	1.06 $\times$	1.48$\times$	1.58$\times$	0.26 $\times$
Geomean								1.32$\times$	1.84$\times$	2.44$\times$	0.24 $\times$

Table 37: Per-pass PAPI counters for `PiecewiseFunctions` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.7e+08 /3.2e+08	5.7e+08 /1.2e+09	1.4e+06 /2.1e+06	1.6e+05 /2.7e+05	9.7e+03 /2.5e+04	8.8e+03 /2.4e+04	4.5e+06/ 1.8e+06
autorefineMaxLevel	F	1.0e+08/ 9.8e+07	2.5e+08 /3.7e+08	2.5e+06/ 2.2e+05	4.3e+02/ 5.3e+01	8.9e+03/ 6.2e+03	4.3e+02/ 5.4e+01	6.1e+06/ 9.5e+05
compressMass	F	9.7e+07/ 5.9e+07	2.5e+08 /2.8e+08	2.1e+06/ 3.1e+04	6.7e+02/ 3.2e+02	1.9e+04/ 5.0e+03	6.8e+02/ 3.2e+02	6.3e+06/ 6.0e+05
diffPW	M	2.8e+08 /3.2e+08	5.9e+08 /1.3e+09	8.0e+05 /1.9e+06	1.6e+05 /2.1e+05	1.1e+04 /2.4e+04	8.5e+03 /2.7e+04	4.5e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.1e+08/ 9.4e+07	3.3e+08 /4.9e+08	2.4e+06/ 2.8e+05	7.8e+02/ 1.4e+02	1.5e+04/ 8.9e+03	7.8e+02/ 1.5e+02	6.3e+06/ 1.5e+06
norm2Estimate	F	1.2e+08/ 9.0e+07	3.4e+08 /4.4e+08	1.2e+06/ 4.2e+05	2.9e+02/ 2.5e+02	1.1e+04/ 9.2e+03	3.0e+02/ 2.5e+02	6.2e+06/ 1.5e+06
pmapCutHistogram	F	1.6e+08/ 6.4e+07	2.8e+08 /3.7e+08	1.7e+06/ 6.3e+04	1.4e+03/ 4.0e+02	1.0e+05/ 7.5e+03	1.4e+03/ 4.0e+02	4.8e+06/ 1.2e+06
truncateTolViolations	F	9.8e+07/ 5.9e+07	2.5e+08 /2.9e+08	2.1e+06/ 3.3e+04	6.7e+02/ 3.3e+02	1.7e+04/ 4.8e+03	6.6e+02/ 3.3e+02	6.3e+06/ 6.2e+05
Total		1.2e+09/1.1e+09	2.8e+09/4.7e+09	1.4e+07/5.1e+06	3.2e+05/4.8e+05	2.0e+05/9.0e+04	2.2e+04/5.3e+04	4.5e+07/1.0e+07

Table 46: Per-pass PAPI counters for `Color0ctree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
paletteEntriesQuantized	F	2.1e+08/ 1.7e+08	6.7e+08 /7.6e+08	5.3e+06/ 1.2e+06	3.3e+02/ 1.4e+02	1.2e+04 /4.0e+04	3.3e+02/ 1.4e+02	9.3e+06/ 1.4e+06
quantizationErrorProxy	F	2.3e+08/ 2.2e+08	9.7e+08 /1.2e+09	4.3e+06/ 8.7e+05	1.3e+02 /2.6e+02	1.5e+04 /3.3e+04	1.3e+02 /2.6e+02	9.2e+06/ 2.8e+06
Total		4.4e+08/3.9e+08	1.6e+09/2.0e+09	9.5e+06/2.1e+06	4.6e+02/4.0e+02	2.7e+04/7.3e+04	4.7e+02/4.0e+02	1.9e+07/4.2e+06

Table 47: Per-pass PAPI counters for `Color0ctree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
paletteEntriesQuantized	F	2.7e+08/−	7.8e+08/−	2.1e+06/−	3.5e+02/−	1.1e+04/−	3.6e+02/−	8.3e+06/−
quantizationErrorProxy	F	3.3e+08/−	1.1e+09/−	1.5e+06/−	4.3e+02/−	4.0e+04/−	4.3e+02/−	7.0e+06/−
Total		6.0e+08/−	1.8e+09/−	3.6e+06/−	7.8e+02/−	5.0e+04/−	7.8e+02/−	1.5e+07/−